IR: MCP4725 0x60 -> LM358 -> 2N4401 -> LED

J8: SCL SDA 3V3 GND, 2.00 mm

JP 2-3 = 100R, JP 2-1 = bypass; both OPEN

J1

MCP4725_IR_0x60

OUT 1
GND 2
SCL 3
SDA 4
3V3 5
GND 6

DAC_IR
GND
SCL_IR
SDA_IR
3V3_IR
GNDI2C_IR

J2

IR DAC OUT / GND

1
2

JP1

SCL_IR
SCL_R_IR
R11
100R 1%
SCL_H_IR

JP2

SDA_IR
SDA_R_IR
R12
100R 1%
SDA_H_IR

J8

I2C IR: SCL SDA 3V3 GND

SCL 1
SDA 2
3V3 3
GND 4

Module ADDR: GND -> 0x60

Pin 2 = board GND; pin 6 = GNDI2C. Same VSS inside module.

5V

J5

IR LED A / K

1
2

LED_K_IR

R1

10k 1%

CMD_IR

U1A

LM358P / DIP8 socket

R3

1k

AMP_IR
BASE_IR

Q1

2N4401 / E-B-C

R2

10k 1%

SENSE_IR

R4

100R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 100 ohm)

C2

DNP loop comp

AMP_IR
SENSE_IR

C2: DNP - DO NOT FIT until loop measured

RED: MCP4725 0x61 -> LM358 -> 2N4401 -> LED

J9: SCL SDA 3V3 GND, 2.00 mm

JP 2-3 = 100R, JP 2-1 = bypass; both OPEN

J3

MCP4725_RED_0x61

OUT 1
GND 2
SCL 3
SDA 4
3V3 5
GND 6

DAC_RED
GND
SCL_RED
SDA_RED
3V3_RED
GNDI2C_RED

J4

RED DAC OUT / GND

1
2

JP3

SCL_RED
SCL_R_RED
R13
100R 1%
SCL_H_RED

JP4

SDA_RED
SDA_R_RED
R14
100R 1%
SDA_H_RED

J9

I2C RED: SCL SDA 3V3 GND

SCL 1
SDA 2
3V3 3
GND 4

Module ADDR: VCC -> 0x61

Pin 2 = board GND; pin 6 = GNDI2C. Same VSS inside module.

5V

J6

RED LED A / K

1
2

LED_K_RED

R6

10k 1%

CMD_RED

U1B

LM358P / DIP8 socket

R8

1k

AMP_RED
BASE_RED

Q2

2N4401 / E-B-C

R7

10k 1%

SENSE_RED

R9

100R 1% 0.25W

GND

I_LED approximately V_DAC / (2 x 100 ohm)

C4

DNP loop comp

AMP_RED
SENSE_RED

C4: DNP - DO NOT FIT until loop measured

SHARED SUPPLY: U1A + U1B + U1C = ONE LM358 DIP-8

J7

5V INPUT / GND

1
2

5V
GND

C7

10uF 16V

C5

100nF X7R 50V

U1C

LM358P / DIP8 socket

V+ 8
V- 4

C5: ceramic, non-polar. C7: electrolytic, + to 5V.

MCP4725 VCC = 3.3 V ONLY; J8 feeds IR, J9 feeds RED.

ADDR is configured on each module, not by this carrier PCB.

H1 M3 / 3.2mm
H2 M3 / 3.2mm
H3 M3 / 3.2mm
H4 M3 / 3.2mm

No shared I2C, 3V3 or I2C-ground copper between the two modules.

OUT pins are DAC outputs: never connect them together.

GNDI2C_x has no PCB copper to GND; joined only inside its module.

For RED: remove the old ADDR-GND bridge before selecting VCC.

70 x 40 mm
Nguyen Nhat Huy - Pham Thanh Vy

Sheet: /
File: led_ir_driver.kicad_sch

Title: DATN: PPG-Simulator - LED / IR driver

Size: A3	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 1/1